Employee Data Analysis Report

This report summarizes the employee dataset after data cleaning, statistical analysis, and visualization.

1. Dataset Overview

Metric	Result
Total employees (rows)	50
Total columns	9
Duplicate rows	0
Missing values	0

2. Statistical Summary

Statistic	Age	Monthly Income
Average	32.62	70,410.00
Median	32.00	69,000.00
Minimum	23.00	39,000.00
Maximum	47.00	99,000.00

3. Key Findings and Patterns

• The department with the most employees is Engineering, with 12 employees.

• The city with the most employees is Delhi, with 7 employees.

• Engineering has the highest average monthly income (82,416.67).

• The correlation between age and monthly income is 0.87. A positive value indicates an upward linear relationship, while a negative value indicates a downward linear relationship.

• The average employee age is 32.6 years.

• The average monthly income is 70,410.00.

4. Data Visualizations

Chart 1: Employee Age Distribution

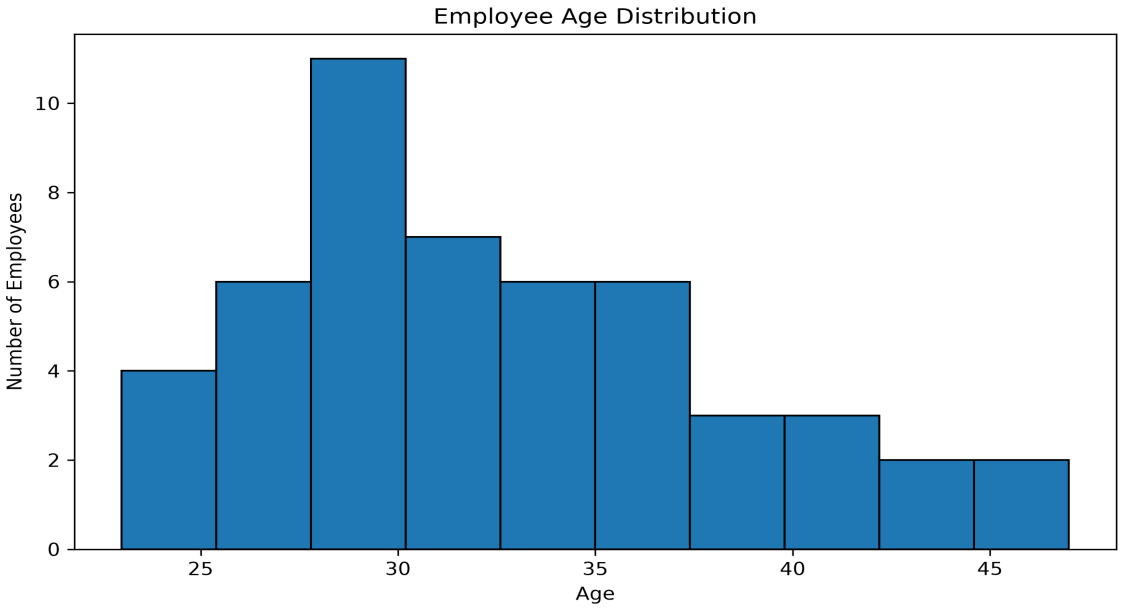


Chart 2: Employees by Department

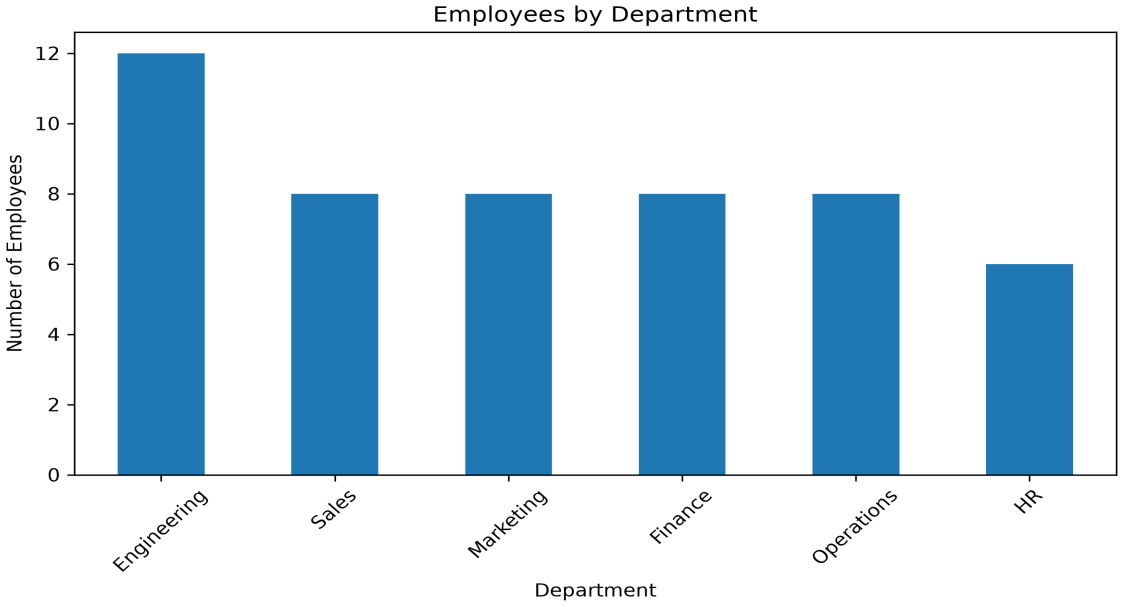


Chart 3: Average Income by Department

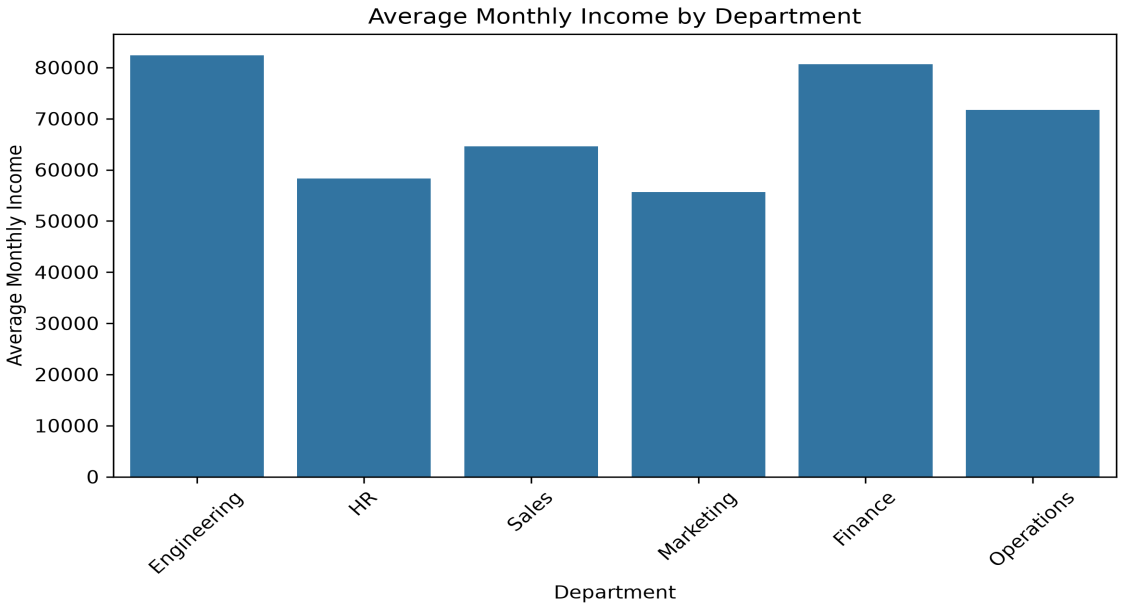


Chart 4: Monthly Income Distribution

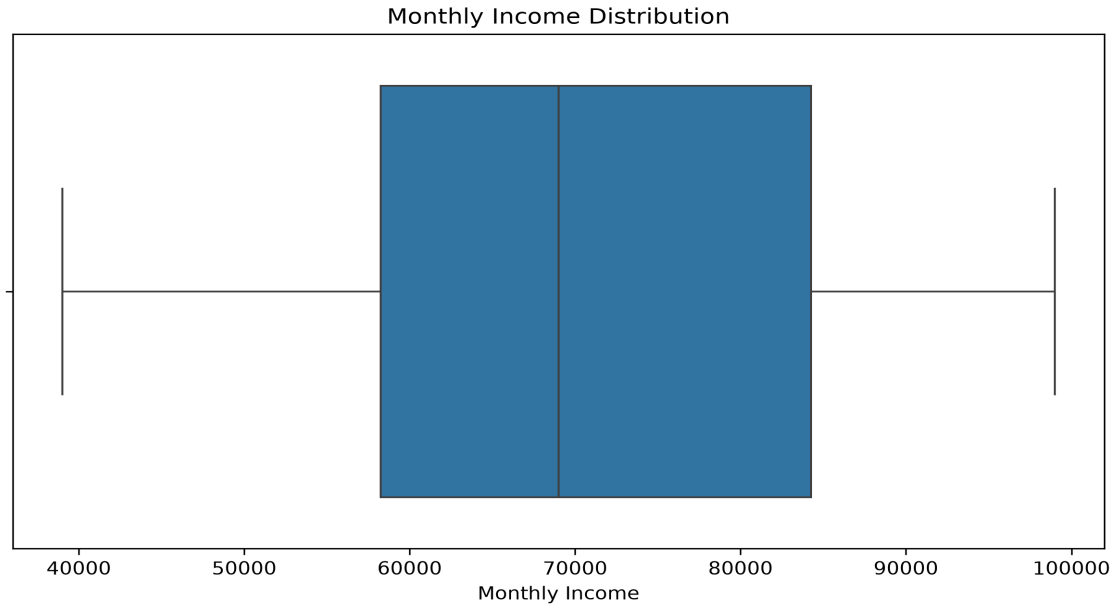
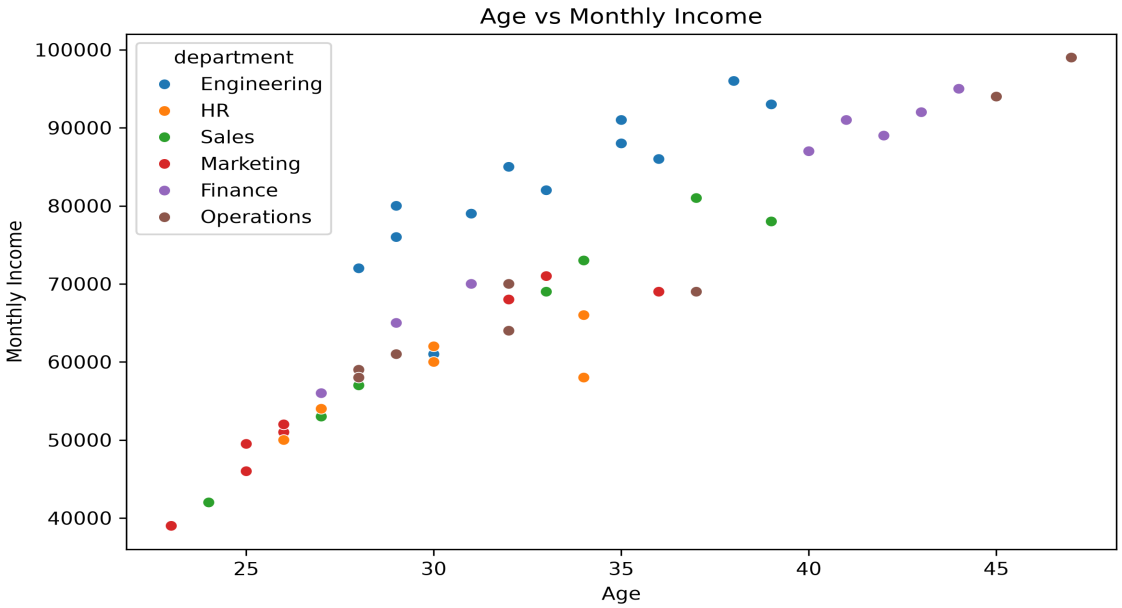


Chart 5: Age vs Monthly Income



5. Conclusion

The employee dataset was cleaned and analyzed using Python, Pandas, Matplotlib, and Seaborn. Statistical measures were calculated, employee patterns were investigated, and five visualizations were prepared. These results help summarize employee demographics, income distribution, and differences between departments and cities.